CLASS 10 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 11: **Areas Related to Circles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The area of a sector of a circle of radius 21 cm is 115.5 cm^2. The central angle of the sector is: 1

(a) 150° **(b)** 15° **(c)** 30° **(d)** 60°

2. A circular field has radius 28 m. Fencing costs ₹ 15 per metre. The cost (in ₹) of fencing the field once is: 1

(a) ₹ 2640 **(b)** ₹ 36960 **(c)** ₹ 176 **(d)** ₹ 1320

3. A wheel has radius 28 cm. The number of complete revolutions it makes to cover 88 m is: 1

(a) 25 **(b)** 100 **(c)** 50 **(d)** 60

4. The radii of two circles are 15 cm and 7 cm. The radius of the circle whose circumference equals the sum of their circumferences is: 1

(a) 44 cm **(b)** 11 cm **(c)** 24 cm **(d)** 22 cm

5. The radii of two circles are 7 cm and 24 cm. The radius of the circle whose area equals the sum of their areas is: 1

(a) 17 cm **(b)** 25 cm **(c)** 27 cm **(d)** 31 cm

6. The minute hand of a clock is 21 cm long. The area swept by the minute hand in 20 minutes is: 1

(a) 44 cm^2 **(b)** 924 cm^2 **(c)** 462 cm^2 **(d)** 1386 cm^2

7. A horse is tied at a corner of a large square field with a rope 28 m long. The area of the field it can graze is: 1

(a) 616 m^2 **(b)** 44 m^2 **(c)** 1232 m^2 **(d)** 2464 m^2

8. The area of a sector of a circle of radius 21 cm is 462 cm^2. The central angle of the sector is: 1

(a) 60° **(b)** 120° **(c)** 240° **(d)** 150°

9. A circular field has radius 42 m. Fencing costs ₹ 8 per metre. The cost (in ₹) of fencing the field once is: 1

(a) ₹ 264 **(b)** ₹ 1056 **(c)** ₹ 2112 **(d)** ₹ 44352

10. A wheel has radius 21 cm. The number of complete revolutions it makes to cover 132 m is: 1

(a) 100 **(b)** 110 **(c)** 50 **(d)** 200

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A horse is tied at a corner of a large square field. The rope is first 7 m long and then lengthened to 21 m. Find the increase in the area it can graze. **2**

12. The minute hand of a clock is 14 cm long. Find the distance moved by its tip in 9 minutes. **2**

13. A wire in the form of a circle of radius 42 cm is bent into a square. Find the area of the square. **2**

14. A horse is tied at a corner of a large square field. The rope is first 14 m long and then lengthened to 28 m. Find the increase in the area it can graze. **2**

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A chord of a circle of radius 35 cm subtends a right angle at the centre. Find the area of the minor segment. **3**

16. A sector of a circle of radius 14 cm has central angle 135°. Find the perimeter of the sector. **3**

17. An umbrella has 4 ribs which are equally spaced and the umbrella, when opened flat, is a circle of radius 14 cm. Find the area between two consecutive ribs. **3**

OR

A chord of a circle of radius 42 cm subtends a right angle at the centre. Find the area of the minor segment. **3**

18. A sector of a circle of radius 14 cm has central angle 225°. Find the perimeter of the sector. **3**

OR

An umbrella has 16 ribs which are equally spaced and the umbrella, when opened flat, is a circle of radius 56 cm. Find the area between two consecutive ribs. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A chord of a circle of radius 21 cm subtends an angle of 120° at the centre. (Use $\pi = \frac{22}{7}$ and $\sqrt{3} = 1.73$; answer to two decimal places.) Find the area of the minor segment. **5**

OR

A square is inscribed in a circle of radius 14 cm. Find the area of the circle outside the square. **5**

20. A chord of a circle of radius 28 cm subtends an angle of 120° at the centre. (Use $\pi = \frac{22}{7}$ and $\sqrt{3} = 1.73$; answer to two decimal places.) Find the area of the minor segment. **5**

OR

A square is inscribed in a circle of radius 21 cm. Find the area of the circle outside the square. **5**

END OF QUESTION PAPER — Total: 40 Marks

Mastermind